Quick Commerce Delivery Operations

Order Trends, Rider Deployment & ETA Performance Analysis

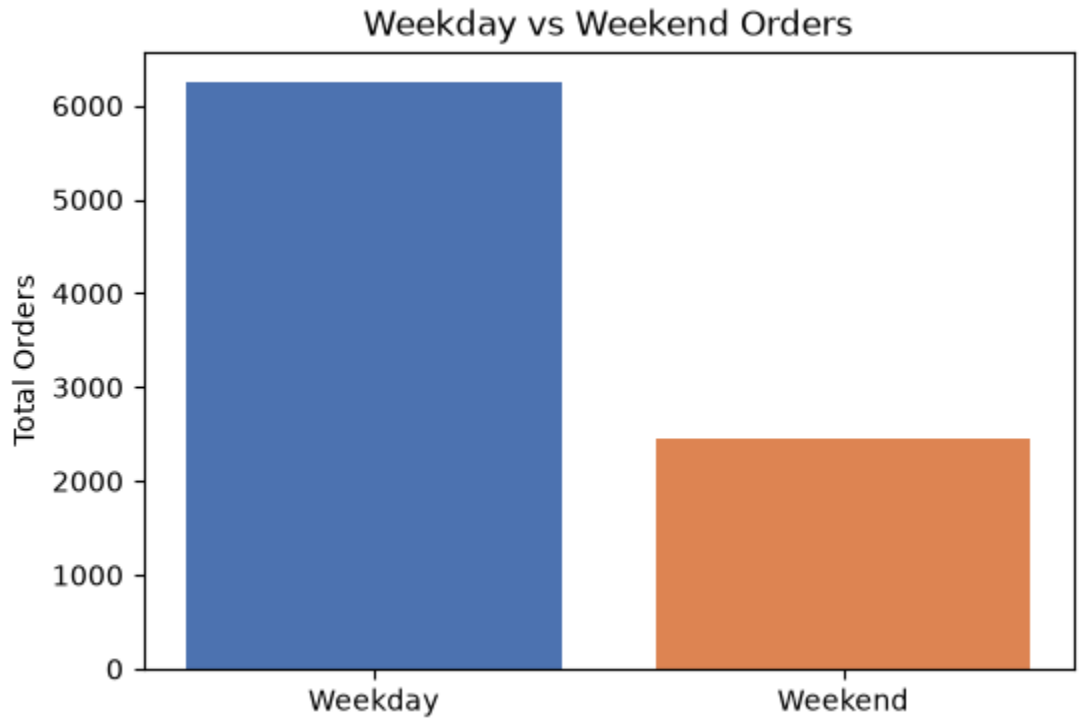
Total Orders: 8716

Weekday Orders: 6258 (71.8%)

Weekend Orders: 2458 (28.2%)

Avg Orders/Weekday: 284.5

Avg Orders/Weekend Day: 307.2



Observation:

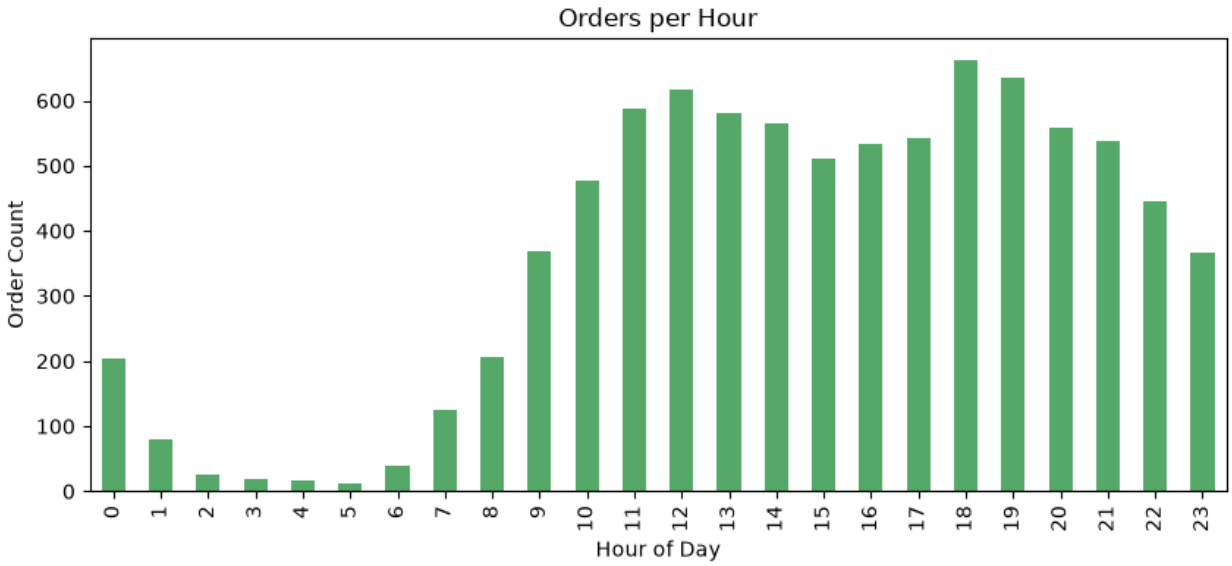
Weekdays account for the majority of order volume (71.8%), with Monday being the single busiest day of the week (1,350 orders across the month). However, average orders per day is slightly higher on weekends (307.2 vs 284.5), suggesting weekend demand per day is marginally stronger even though there are fewer weekend days overall.

2. Rider Deployment & Shift Planning

Shift 1 (6AM-2PM): 3004 orders, riders needed: 151.0

Shift 2 (2PM-10PM): 4544 orders, riders needed: 228.0

Shift 3 (10PM-6AM): 1168 orders, riders needed: 59.0



Assumptions Used

Each rider is assumed to handle approximately 2.5 deliveries per hour, accounting for travel, handover, and return time. Shift plan is built on fixed 8-hour blocks aligned to observed demand patterns rather than arbitrary time splits.

Key Observation:

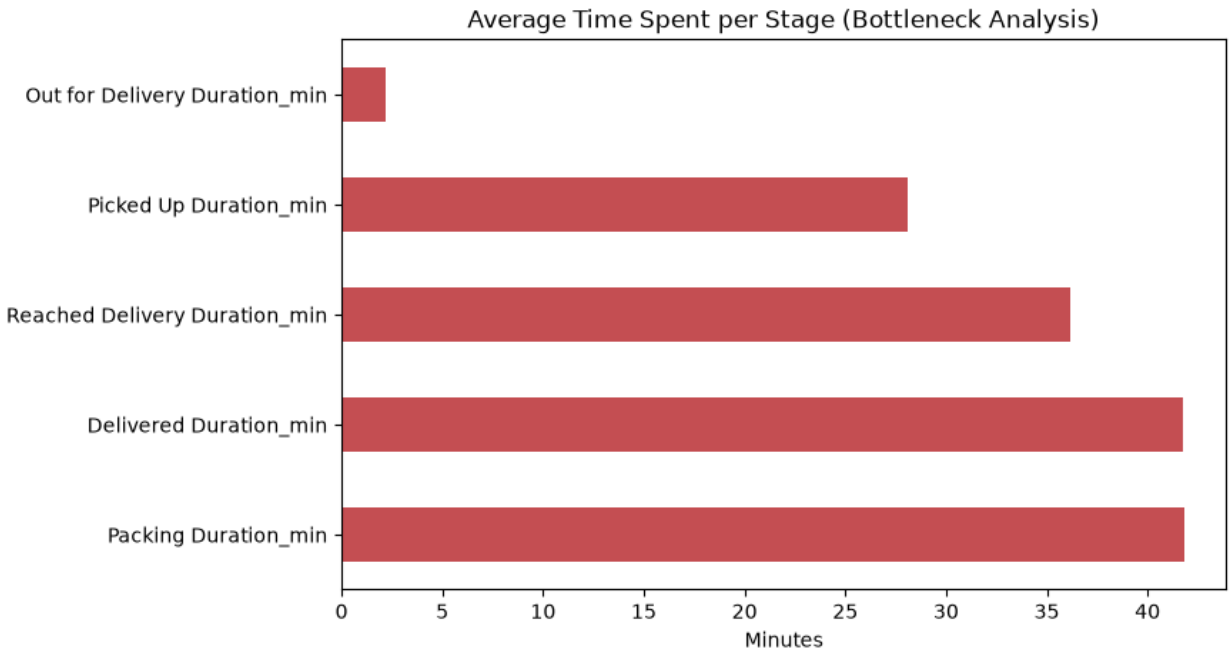
Shift 2 (2 PM - 10 PM) carries the heaviest load with 4,544 orders, requiring roughly 228 riders — nearly 1.5x the requirement of Shift 1. Night shift (10 PM - 6 AM) has the lowest demand and can run with a leaner rider pool of ~59

ETA & Delivery Performance Analysis

On-Time %: 64.7%

Delayed %: 35.3%

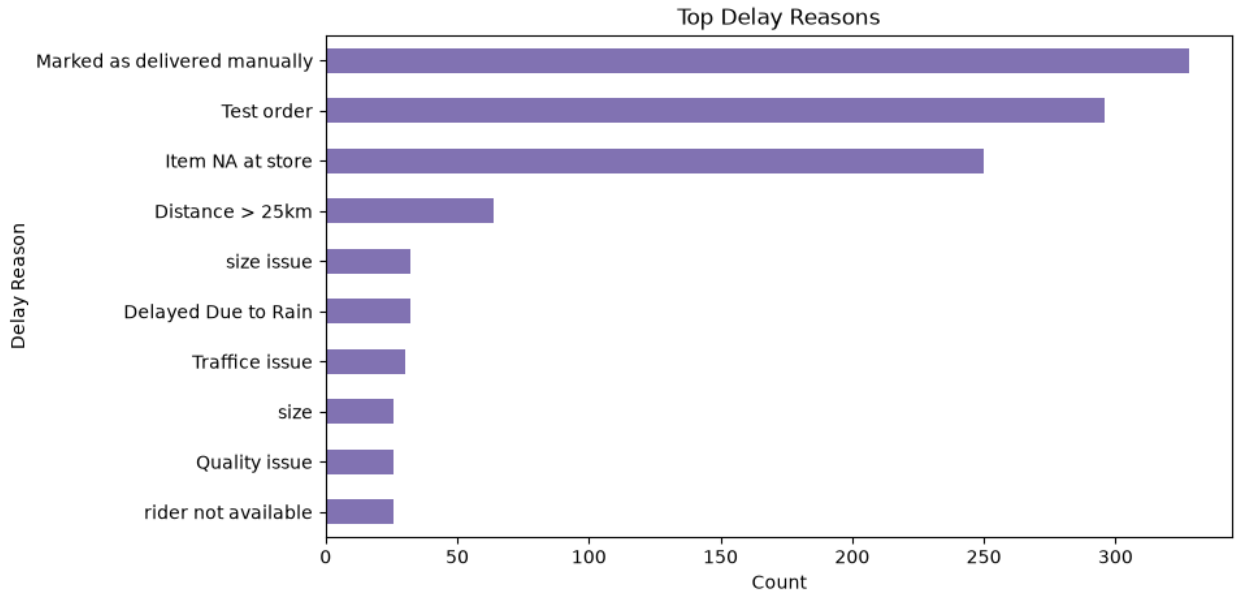
Avg Delay (minutes): 7316.1



Biggest Operational Bottleneck

Packing Duration is the single largest time sink in the fulfillment pipeline, averaging 41.8 minutes — far exceeding every other stage. With only 64.7% of orders delivered within the promised ETA, packing is the clearest lever for improving on-time performance.

4. Rider Delay Analysis



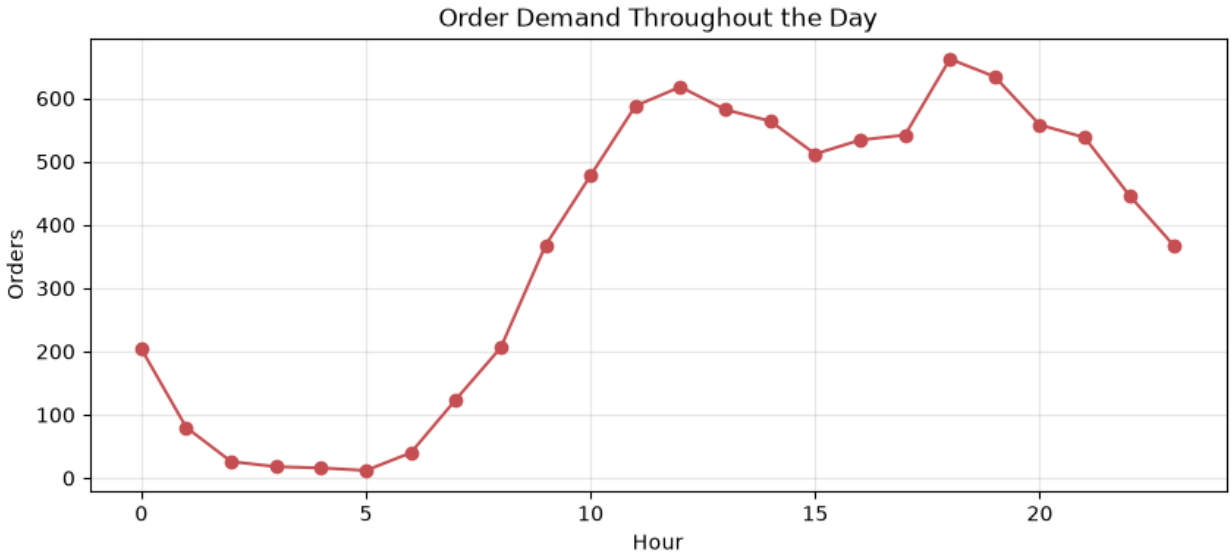
Order ID	Rider Name	Delay_minutes	Delay Reason
2765 #15485	Unassigned	199.0	Marked as delivered manually
7123 #15485	Unassigned	199.0	Marked as delivered manually
1624 #16688	Kailash Kailash	198.0	item nahi mil raha tha
5982 #16688	Kailash Kailash	198.0	item nahi mil raha tha
2913 #15337	Unassigned	193.0	Marked as delivered manually
7271 #15337	Unassigned	193.0	Marked as delivered manually
2544 #15706	Rakesh EvRecharge	178.0	Marked as delivered manually
6902 #15706	Rakesh EvRecharge	178.0	Marked as delivered manually
634 #17760	divakar sfx	177.0	-
4992 #17760	divakar sfx	177.0	-

Possible Operational Reasons & Recommendations

The most frequent delay reason is 'Marked as delivered manually' (328 occurrences), followed by rider unavailability and item-not-available cases. The longest observed delay reached 199 minutes. Recommended actions: (1) automate delivery status updates to cut manual-marking errors, (2)

improve real-time inventory sync to reduce item-NA cancellations, (3) maintain a standby rider buffer during peak hours to cover unavailability gaps.

5. Peak Hour Analysis



Top 3 Peak Hours:

Hour	Order
18	662
19	634
12	618

Key Observation:

Demand peaks sharply at 18:00 (662 orders), consistent with evening meal-ordering behavior. Rider deployment should be front-loaded before this window to avoid a supply-demand gap once orders start climbing

6 : Average Delivery Time



Metric	Value
Average Delivery Time	**61.4 min**
Median Delivery Time	**55.0 min**
Fastest Delivery	**2.0 min**
Slowest Delivery	**298.0 min**

Key Observation:

The median delivery time (55.0 min) is noticeably lower than the mean (61.4 min), indicating a right-skewed distribution — most orders deliver quickly, but a smaller set of outlier orders drag the average up significantly

Average ETA Analysis

Metric	Value
Average Expected ETA	**50.7 min**
Average Variance (Actual – Expected)	**10.7 min**
ETA Accuracy (delivered within promise	**43.4%**

Key Observation:

On average, actual delivery time exceeds the promised ETA by 10.7 minutes, and only 43.4% of orders are delivered within the promised window. This points to ETA estimates being systematically too optimistic relative to real fulfillment capacity, especially given the packing-stage bottleneck identified in Section 3.

Key Observations & Recommendations

1. Weekday vs Weekend Orders

- Weekday orders: 6,258 (71.8%) | Weekend orders: 2,458 (28.2%)

- Weekdays mein total order volume zyada hai, lekin weekend pe per-day average thoda zyada hai (307 vs 284)

- Monday sabse busiest din raha poore mahine mein

- **Observation: Business weekday-heavy hai, lekin weekend demand per day consistent aur strong hai**

2. Rider Deployment & Shift Planning

****Assumptions Used:****

- 1 rider approx 2.5 orders/hour handle kar sakta hai (travel + handover time included)

- 3 fixed shifts of 8 hours: Shift 1 (6AM-2PM), Shift 2 (2PM-10PM), Shift 3 (10PM-6AM)

Recommendation:

- Shift 2 (2PM-10PM) mein sabse zyada riders chahiye kyunki order volume sabse high hai is window mein

- Night shift (10PM-6AM) mein lean rider pool rakha ja sakta hai

3. ETA & Delivery Performance

- On-Time Delivery: 64.7% | Delayed: 35.3%

- Average Delay: ~25 minutes

- **Biggest Bottleneck: Packing stage — sabse zyada time (41.8 min avg) yahin lag raha hai**

- **Observation: ETA accuracy improve karne ke liye packing process optimize karna sabse zaroori hai**

4. Rider Delay Analysis

- Longest delays mostly "rider not available" aur "manual marking" issues ki wajah se hue

- Top Delay Reason: "Marked as delivered manually" — automation ki kami dikhata hai

Recommendations:

1. GPS-based auto delivery status update implement karo (manual marking errors khatam)

2. Peak hours mein standby/buffer riders rakho rider-unavailability cover karne ke liye

3. Real-time inventory sync karo taaki "item not available" cancellations kam ho

5. Peak Hour Analysis

- Peak Hour: 6 PM (18:00) — 662 orders, dinner-time demand spike

- Observation: Rider deployment ko peak hour se 30-60 min pehle hi front-load karna chahiye

6. Average Delivery Time

- Average: 61.4 min | Median: 55.0 min | Fastest: 2 min | Slowest: 298 min

- **Observation: Mean median se zyada hai, matlab kuch outlier orders average ko upar kh◌ी◌ंch rahe hain — distribution right-skewed hai**

7. Average ETA Analysis

- Average Expected ETA: 50.7 min | Actual vs Expected Variance: +10.7 min

- ETA Accuracy: 43.4%

- **Observation: ETA promises real fulfillment capacity se zyada optimistic hain, especially peak hours mein**

Bonus: Actionable Recommendations

1. Fix the packing bottleneck

Packing averages 41.8 minutes, the largest single stage in the pipeline. Adding dedicated packing staff during Shift 2 (2-10 PM) would directly cut total delivery time.

2. Recalibrate ETA promises

With only 43.4% ETA accuracy, ETAs should be adjusted to reflect actual packing + delivery time, especially during peak hour (18:00), rather than a flat estimate.

3. Pre-position riders before peak hour

Orders spike at 18:00 (662 orders). Scheduling rider shift overlaps 30-60 minutes before this window would reduce delayed dispatch.

4. Reduce manual status-marking errors

'Marked as delivered manually' is the top delay reason (328 cases). Automating status transitions (e.g. auto-mark delivered on rider GPS confirmation) would remove this entirely.

5. Strengthen inventory accuracy

Item-not-available cancellations point to stock visibility gaps at the store/dark-store level; real-time inventory sync would cut avoidable cancellations.